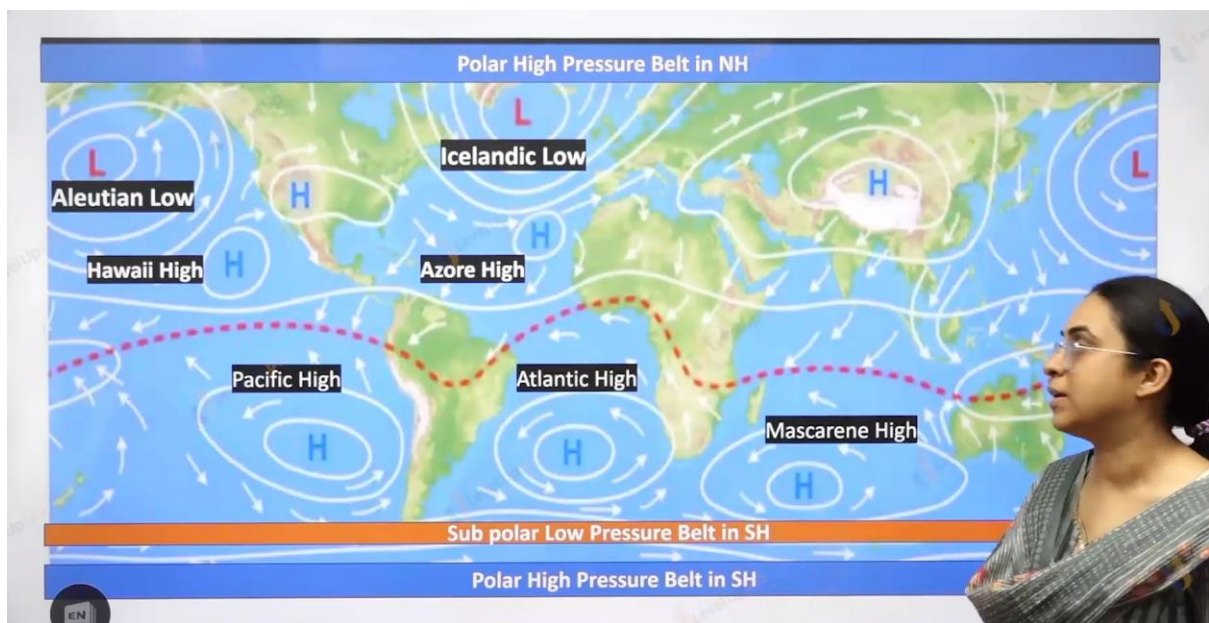
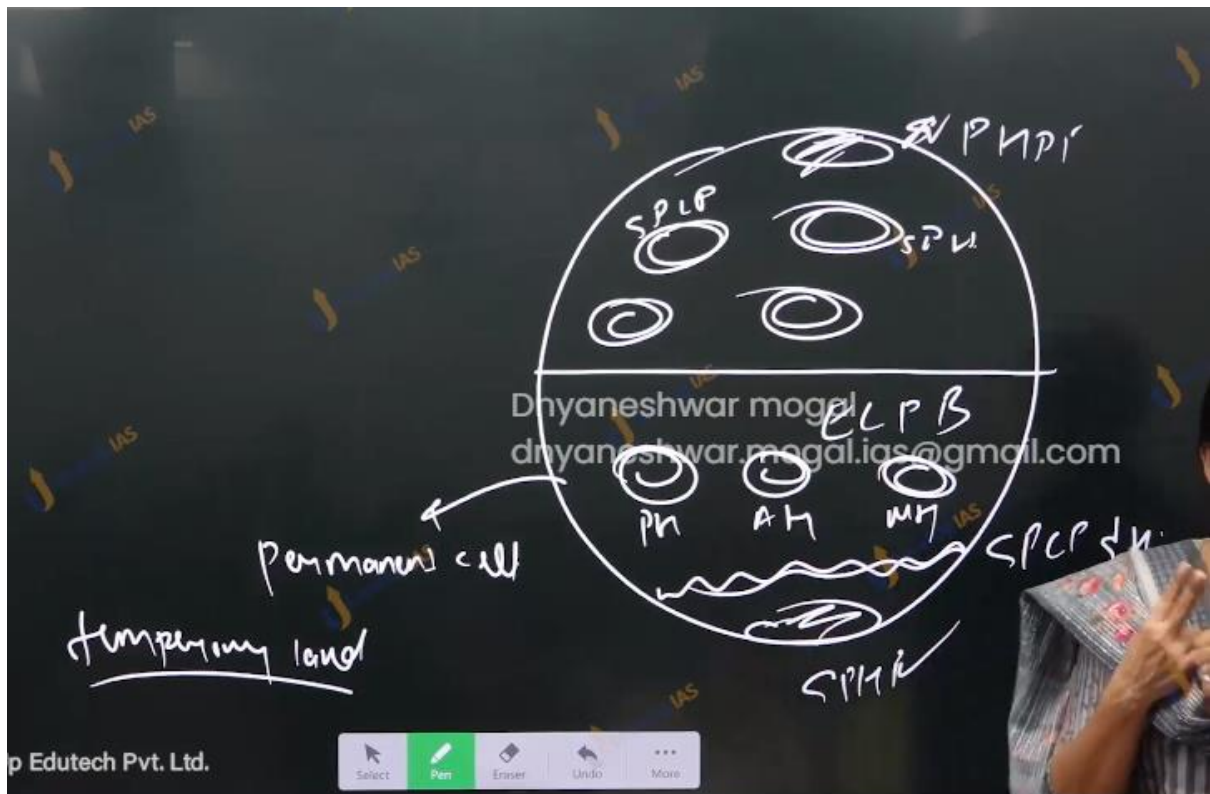


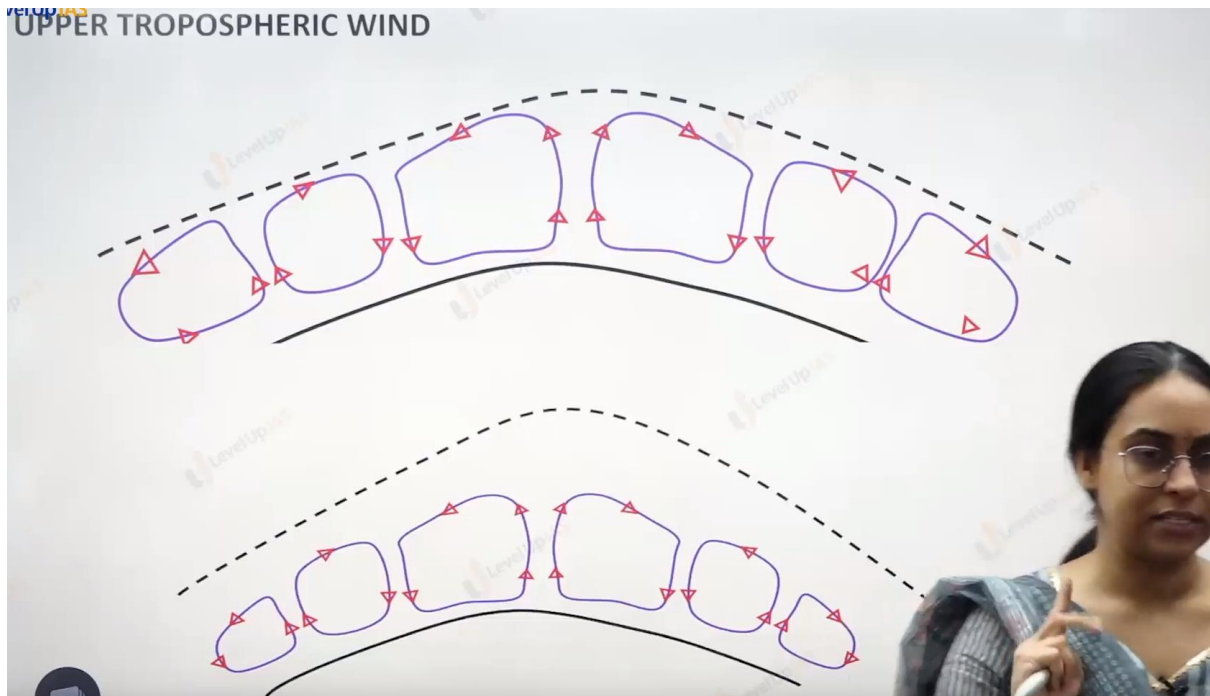
## (20) Air mass & Front

### 1. AIR MASS

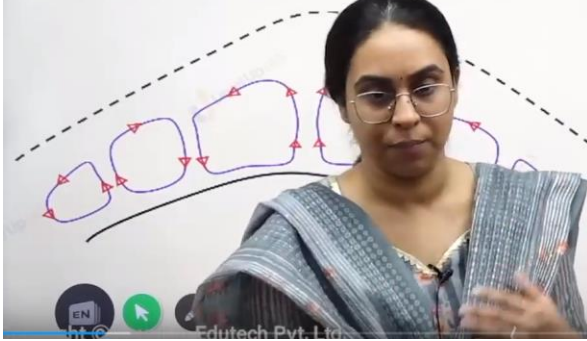
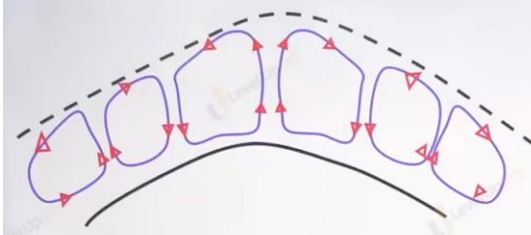




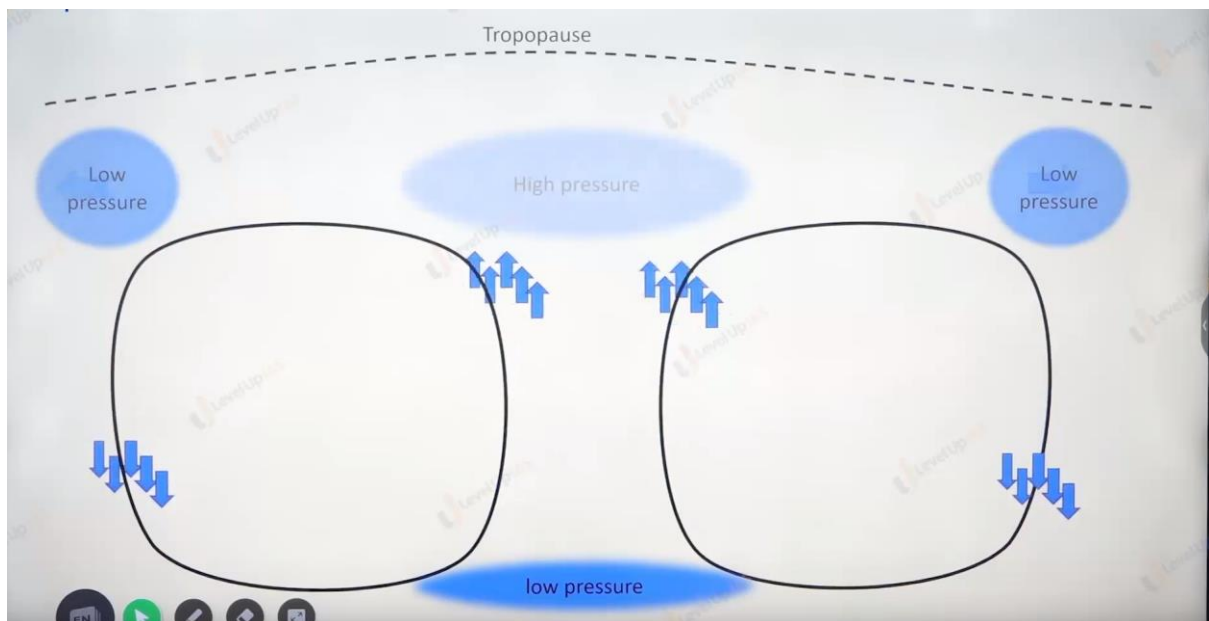
## UPPER TROPOSPHERIC WIND



## UPPER TROPOSPHERIC WIND



- As per Ferrel, UT winds are mirror oopposite of surface winds but in reality the cells are shallow. Upper troposphere has a different conditions.
- Along the EQ at upper troposphere there is a high pressure while along the polar regions at the upper troposphere there is low pressure. PGF is directed away from the equator to the poles.
- In the upper troposphere, wind speed is very high as winds experience very minute friction which is considered negligible.
- Wind is majorly governed by PGF and Coriolis force. When the wind is completely balanced by two forces, it is called as GEOSTROPHIC WIND.
- UT winds are majorly westerly winds. (Move from West to East)



## Rossby Wave

- Meandering Rossby wave create additional pressure pattern in closed loop cells of alternate High Pressure and Low Pressure in the upper troposphere
- The pressure pattern on upper troposphere creates additional pressure pattern on Earth's surface
- Rossby wave can go through a life cycle of 10-15 days where they change from near latitudinal west to east straight line to meandering north to South Path: INDEX CYCLE OF ROSSBY WAVES.

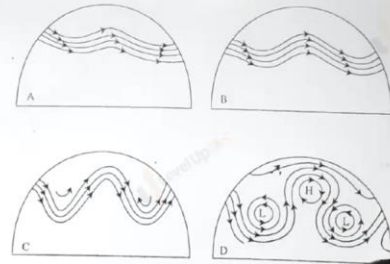


Fig. 6.10 : Index cycle of jet streams.



### British Weather

9:00am

9:01am

9:02am

9:03am

**WEATHER IS UNPREDICTABLE  
DON'T GET CAUGHT OFF-GUARD  
EVER AGAIN**







## Jet Stream

<https://scijinks.gov/jet-stream/>

- Velocity of Jet stream increases because of temperature difference between the cells which provides an additional force which increases speed of Jet stream.
- Some Important Jet Stream are: Sub Tropical Westerly Jet Stream, Polar Jet Stream, Polar Night Jet Stream, Tropical Easterly Jet Stream.

The diagram shows a cross-section of the Earth with a large blue 'H' (High pressure) over the Arctic region and a red 'L' (Low pressure) over the equatorial region. Arrows labeled 'JETSTREAM' show the flow of air from the high to the low pressure area. A text box at the top says 'PREVAILING JETSTREAM' and 'IS ENHANCED BY ARCTIC AMPLIFICATION'. The source 'clideo.com' is noted at the bottom right.

## Sub Tropical Westerly Jet Stream

- It is found around 30-35 degree N/S
- Plays an imp role in bring Western Disturbances in India
- Plays an imp role in burst of Indian Monsoon s
- It is formed at the interface of Hadley and Ferrel Cell

## Polar Jet Stream

- It is found at interface of Ferrel Cell and Polar Cell
- It is found around 65 degree N/S
- Polar Jet Stream experiences prominent meandering and goes through the life cycle of 10-15 days
- It plays an imp role in formation of temperate cyclone and affects the weather of mid latitude

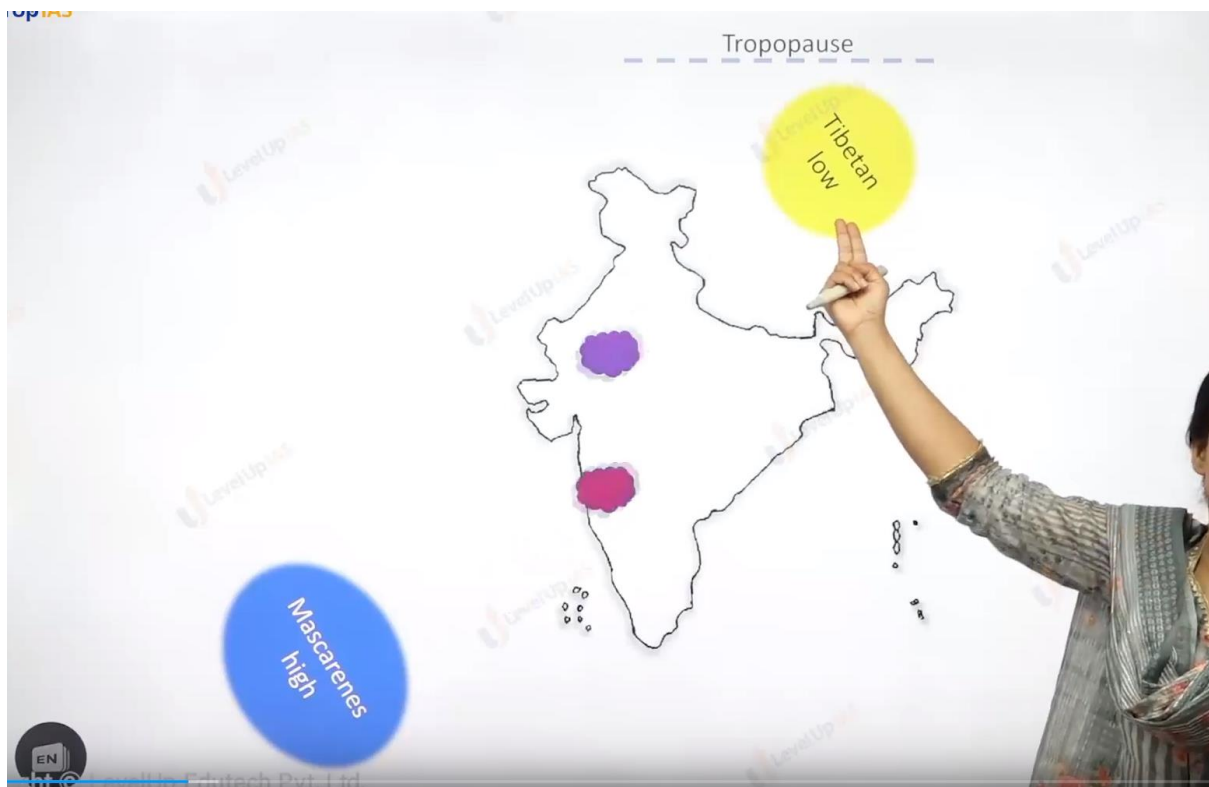
# Polar Night Jet Stream, Tropical Easterly Jet stream

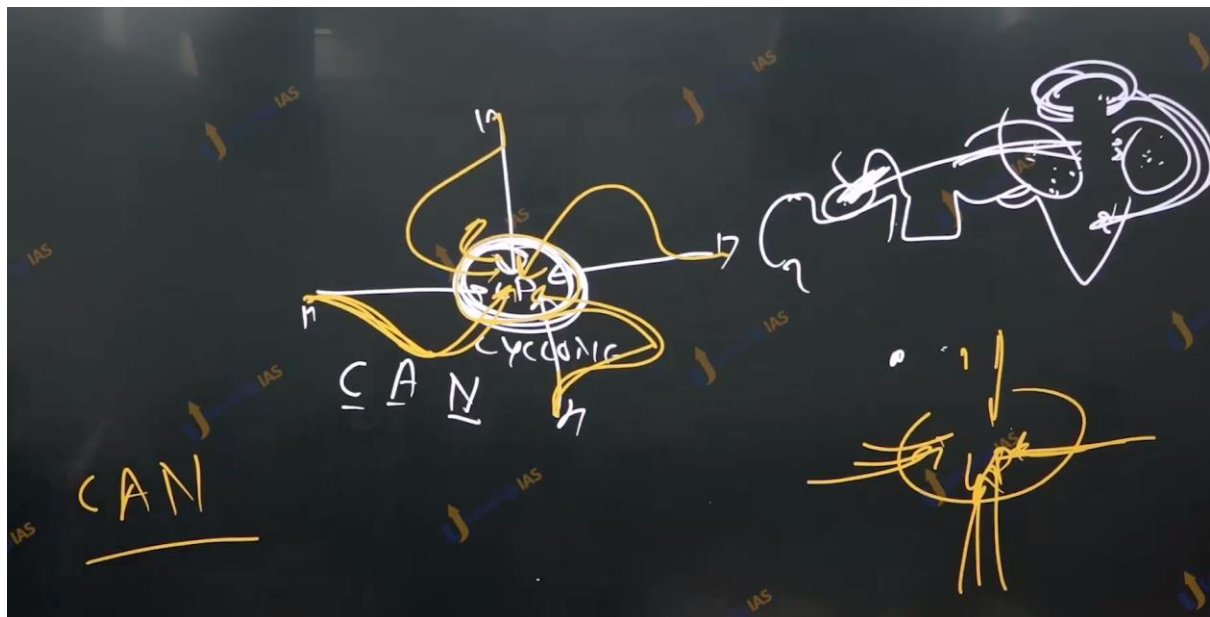
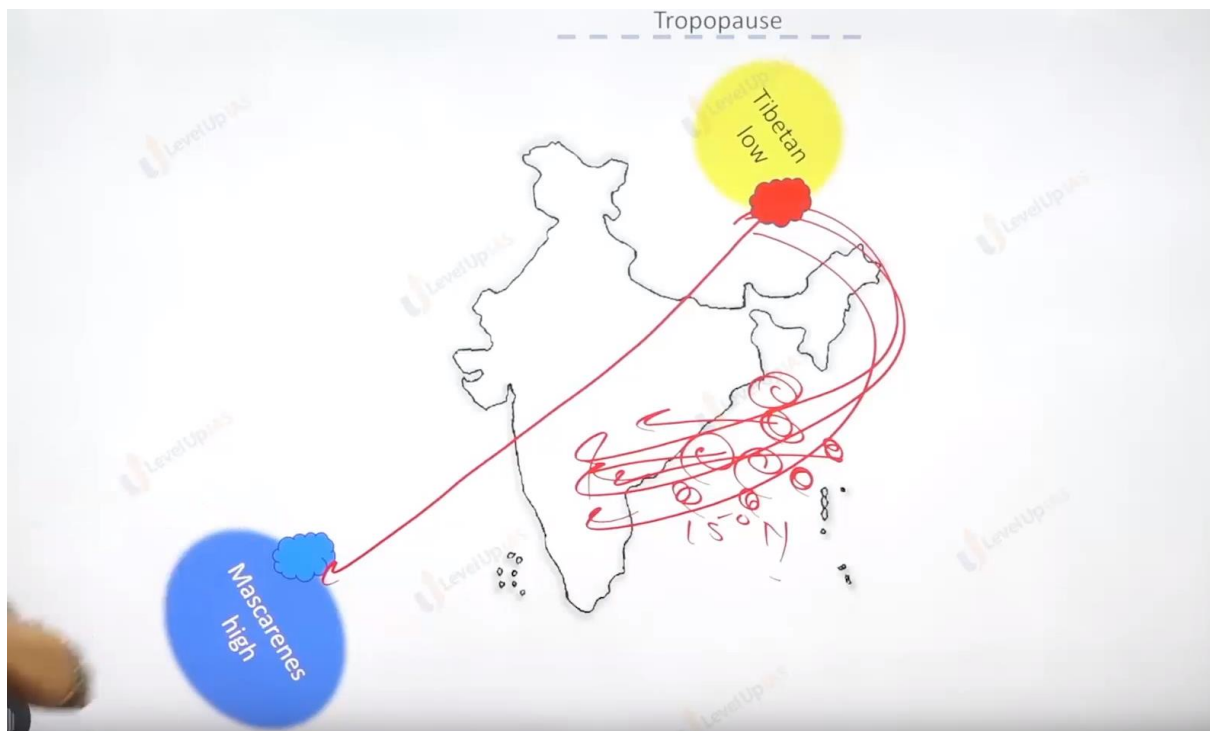
## Polar Night Jet Stream

- It is found at Poles
- It is very powerful in winters

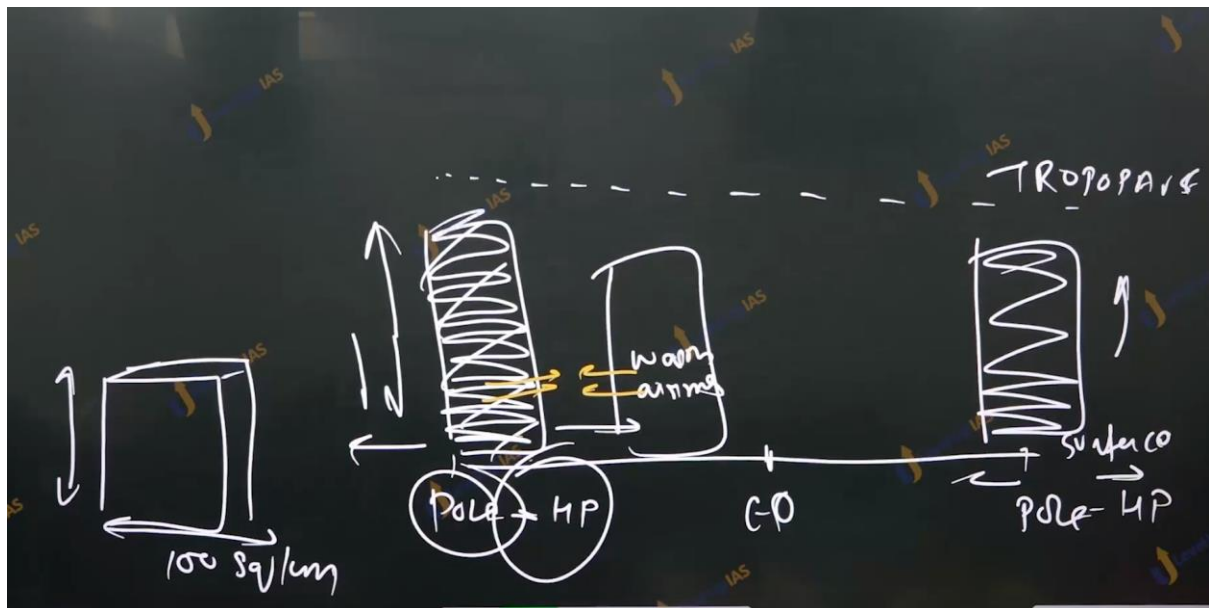
## Tropical Easterly Jet Stream

- Regional/Local Jet Stream. Formed due to heating of Tibetan Plateau
- Affects South Asian Monsoon and distribution of South Asian Monsoon









Cyclone:

1. To understand the temperate cyclone, it is important to understand the concept of air mass.
2. Air mass are massive bodies of air covering 1000 sqkm (horizontally on earth's surface and they extents up to trope pause.)
3. Air masses have nearly same temperature and moisture level horizontally the nature and properties of air mass are acquired from the surfaces over which air mass developed.
4. Air masses formed over relatively flat surfaces and extensive surfaces
5. Air mass develops in the high-pressure conditions as these conditions prevent from air moving ex. Siberian high pressure conditions in winters and Canadian high-pressure conditions.
6. The places for the formation of air mass are polar regions, artic Antarctic, and tropical region.
7. Based on place of formation there are 4 types of air mass.
  - a. Antarctic: this air masses are found in the Antarctic and very cold

- b. Arctic air mass: they are formed in the arctic region, and they are cold.
  - c. Tropical air mass: these air masses formed in the lower lying region and they are comparatively warm.
  - d. Polar: these air masses form high latitude region, and they are cold.
- 

---

8. Based on the nature of surface there can be 2 types of air mass.

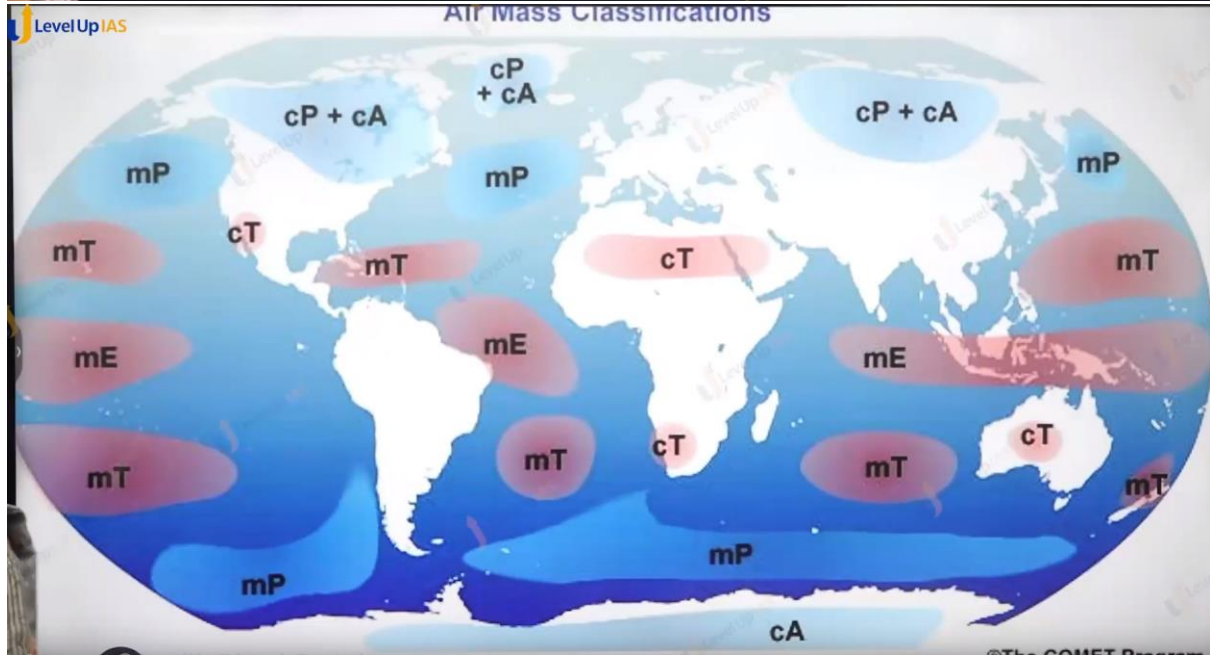
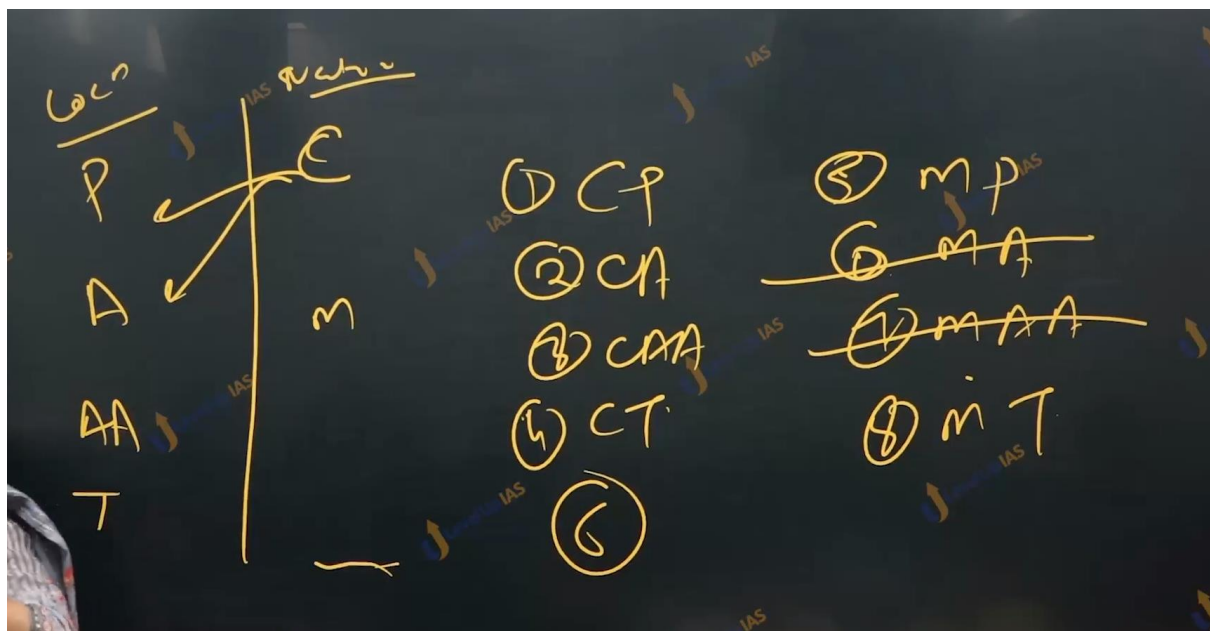
---

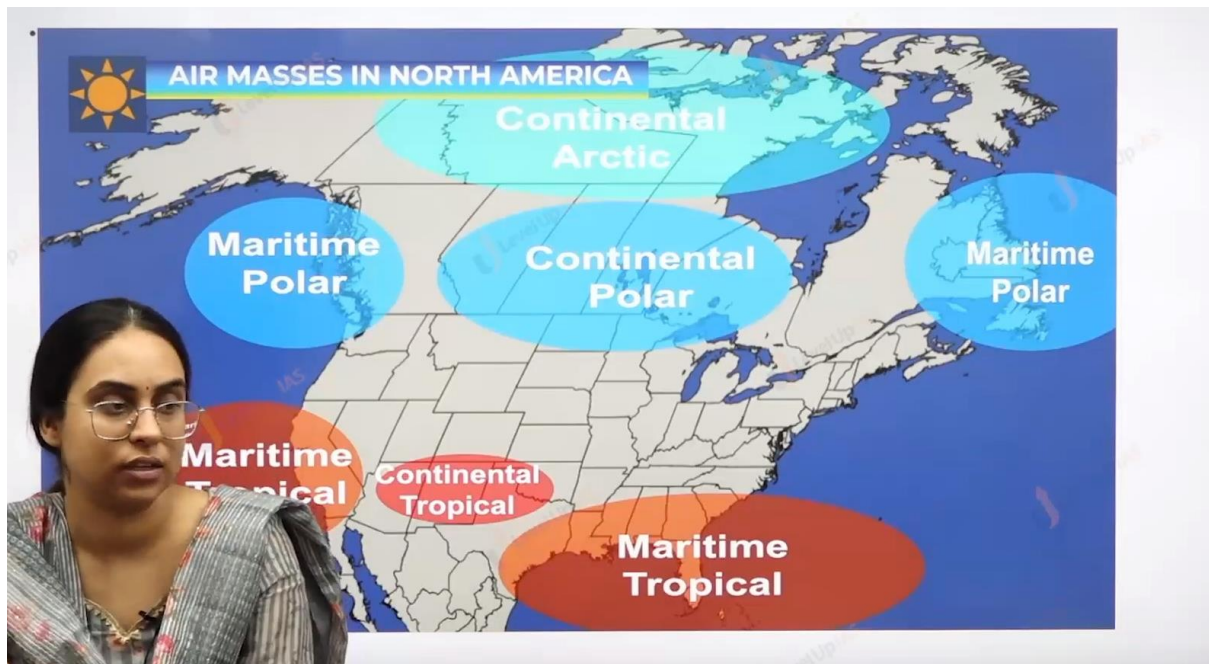
- a. Maritime: they are founded over the water bodies and filled with moisture
  - b. Continental air mass: these are formed over land and therefore they are dry.
- 

9. according 6 type of air mass identified

---

- a. continental arctic.
  - b. Continental Antarctic
  - c. Continental polar Siberian air mass
  - d. Continental tropic ex Sahar
  - e. Maritime tropic ex along Gulf of Mexico
  - f. Maritime polar : ex Continental air mass when crosses the and reaches the ocean it becomes maritime polar.
-





**LevelUp IAS**

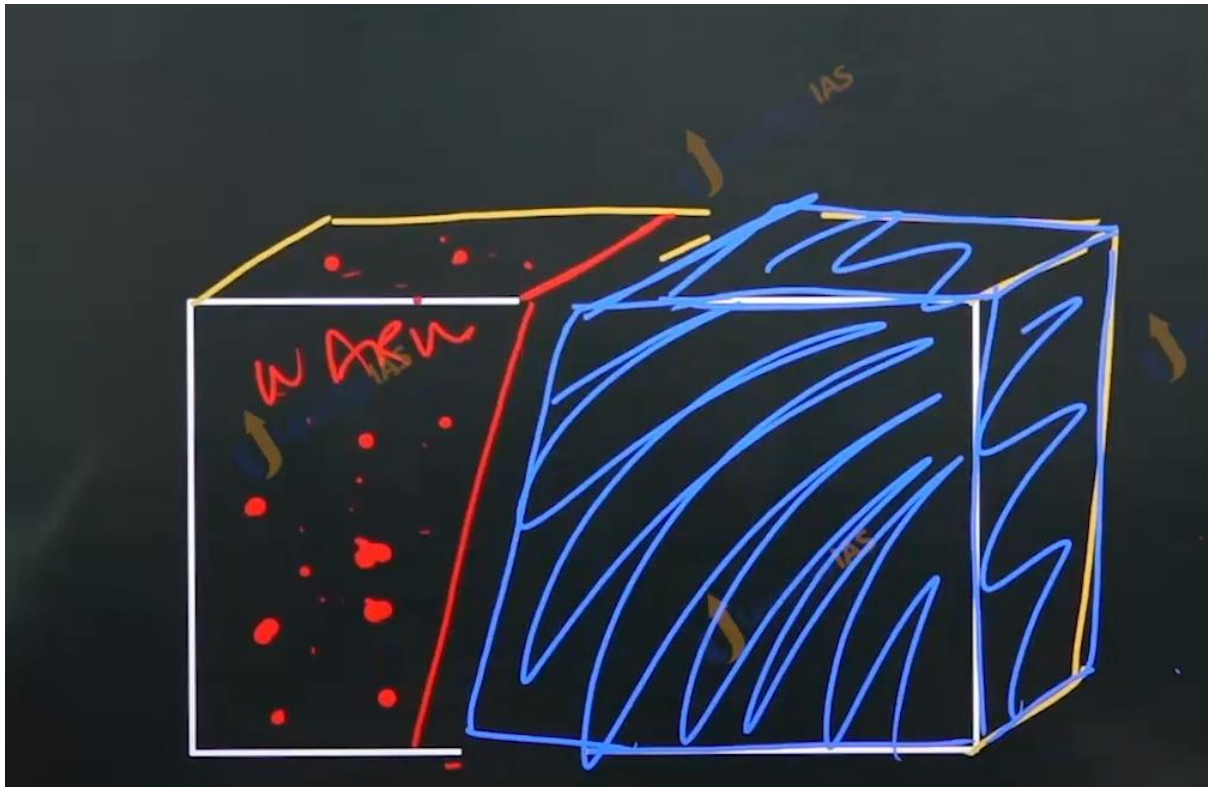
## Fronts

- It is an inclined zone along which two opposing air masses (having contrasting characteristics in terms of air temperature, humidity, density) converge together.
- The frontal zone is inclined at a low angle to the Earth's surface.
- Along a front, one airmass will be relatively warmer than the other airmass. Warmer Airmass will try to rise along the front and cooler airmass will drag beneath.

The diagram illustrates a frontal system in three dimensions. A red block represents a warmer airmass, and a blue block represents a cooler airmass. They are shown meeting at an inclined interface, which is the frontal zone. Arrows indicate the direction of movement and the rising of the warmer air mass along the front. The text 'heshwar.m@gmail.com' is visible on the diagram.

A woman is visible in the bottom right corner of the frame, pointing towards the diagram.





Fronts:

- 
1. It refers to the inclined zone where two opposing (having contrasted the temperature, density, humidity) converge together.
  2. The frontal zone is inclined at a low angle to the surface along the front one air mass will be relatively warmer than the other.
  3. Warmer air mass will try to rise along the front and the cooler air mass will drag beneath the front.
-

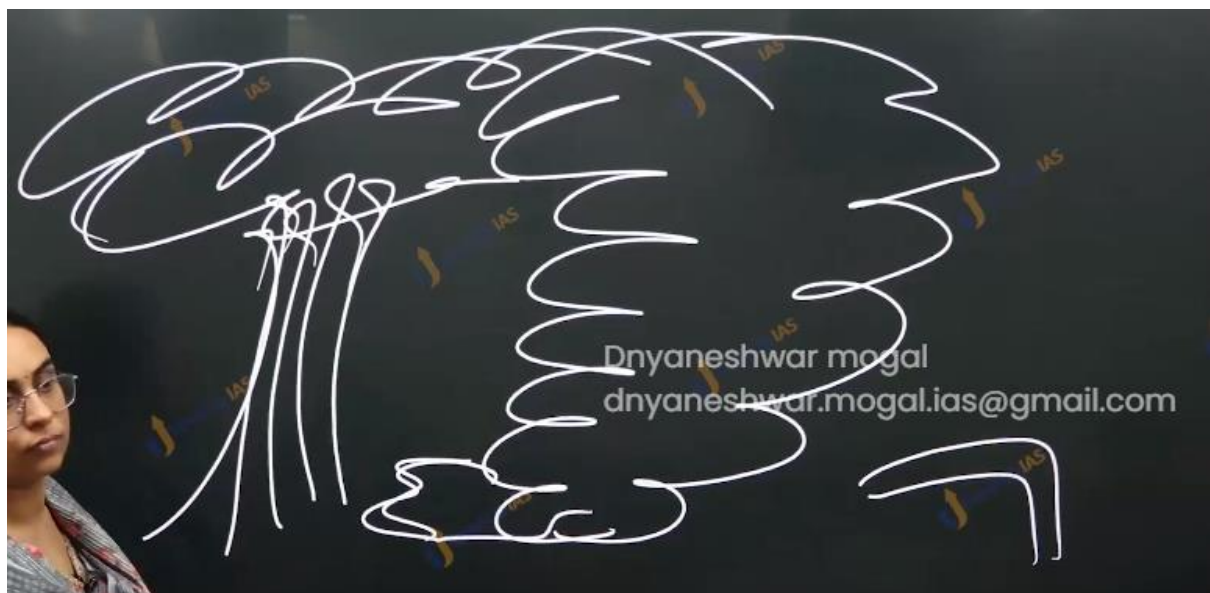
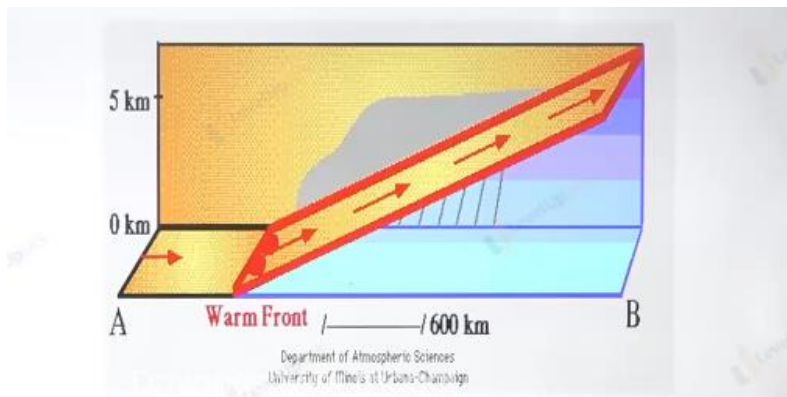
**Types of Fronts**

Based on the nature of convergence, relative acceleration of air masses, and stages of front formation, four types are identified:

- 1. Stationary Front:** 2 airmass standing in front of Each Other
- 2. Warm Front:** Warm airmass is active and moves into region of cold airmass
- 3. Cold Front:** Cold airmass is active and moves into region of warm airmass
- 4. Occluded Front:** Warm airmass is completely

### Type of fronts:

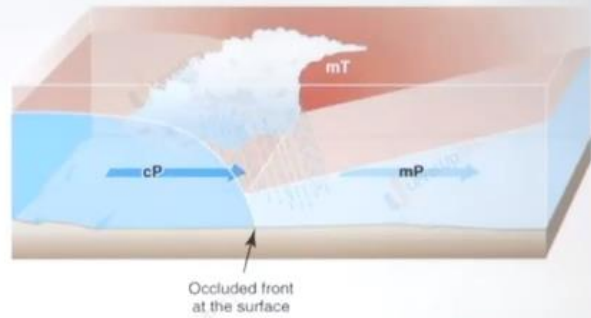
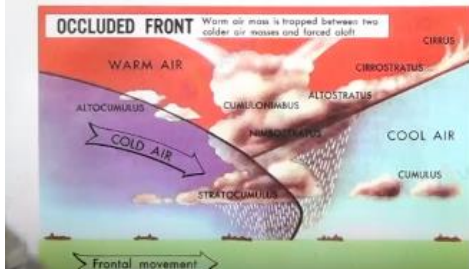
1. Based on the nature of conversion and relative acceleration of the air masses and the stages of front formation 4 type of fronts are identified.
  - a. Stationary fronts: two air masses are standing in front of each other.
  - b. Warm front: warm air mass is active and moved into the region of cold air mass.
  - c. Cold front:
    - i. Cold air mass is active and moved into the region of warm air mass.
    - ii. Cloud with very high vertical extents columbiums air cloud.
    - iii. Columbiums cloud formed along the cold front
  - d. Occluded front: Warm air mass completely tossed above the cold air mass



(columbiums cloud)

## Occluded Front

- When warm air mass is completely raised over the cold air mass it is called an occluded front.
- Weather along this sector is complex—a mixture of cold front type and warm front type weather. These are common in western Europe.

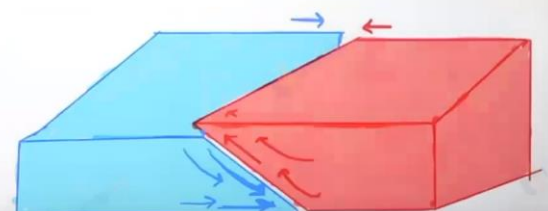
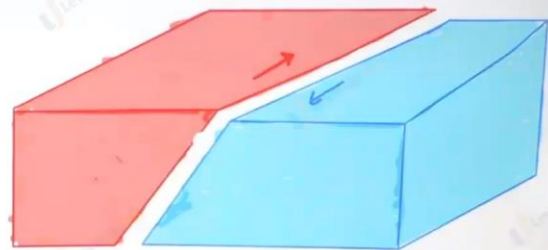


### Nature of fronts:

1. The cold front is relatively steeper and warmer front relatively gentle.
2. Cold air mass moves faster as compared to warm air mass.
3. Warm air mass raises above and cold air mass always pushes from the below whether in the warm front or cold front.

## Stationary Front

- In this type two air masses having contrasting physical properties rest against each other without showing any major movement.
- Weather generally remains stable but sometimes it might rain.

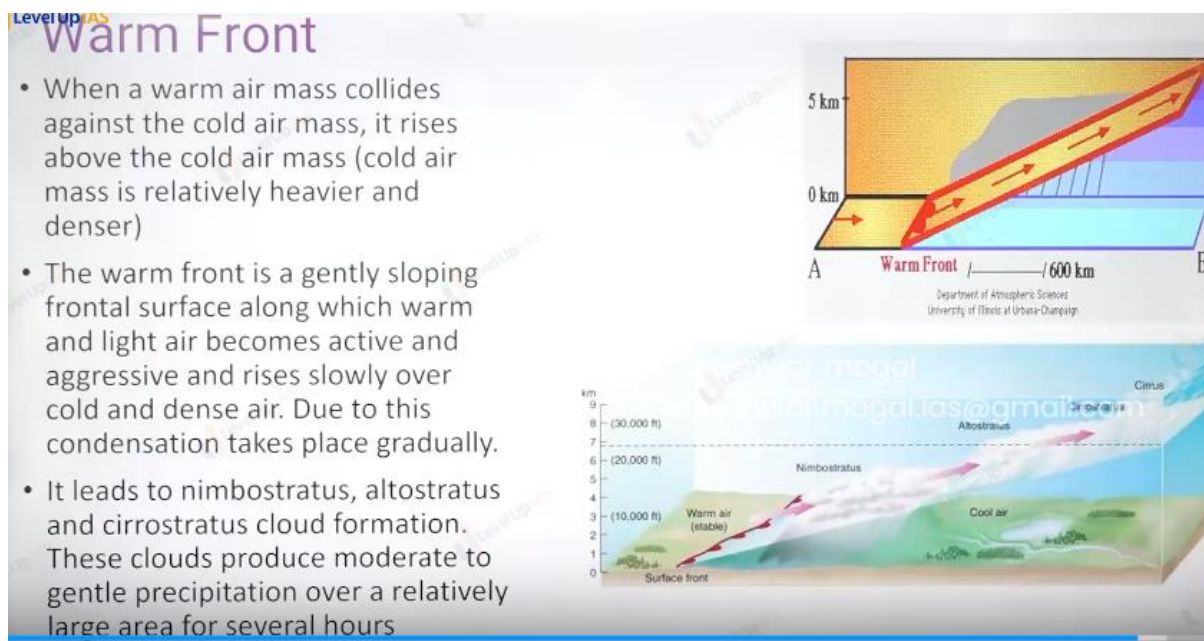




## Stationary front:

---

1. Warm air mass and cold air mass lays side by side without showing any major movement
  2. The weather generally remains stable without any precipitation.
  - 3.
- 



## Warm Front:

---

1. When the warm air mass collides against the cold air mass, it rises above the cold air mass
  2. A warm front is a gently sloping frontal surface along which warm and light air rises. Due to raising of warm air condensation takes place.
  3. It leads to formation of various clouds like nimbostratus, altostratus, cirrostratus, and cirrus.
-

